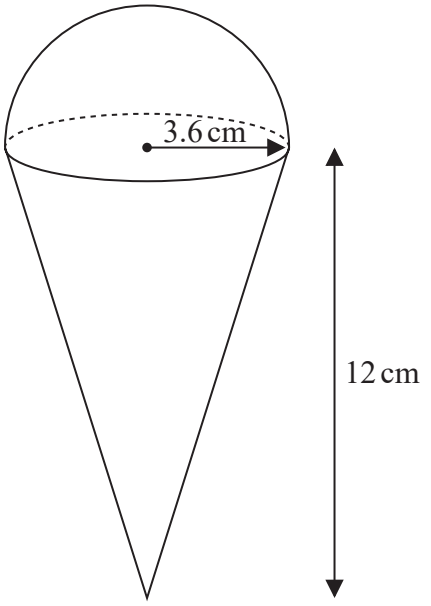
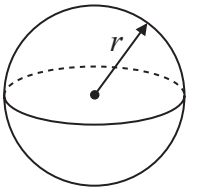
16 Jeff makes two model ice creams, **A** and **B**.

Model **A** is made of a solid hemisphere on top of a solid cone as shown below.



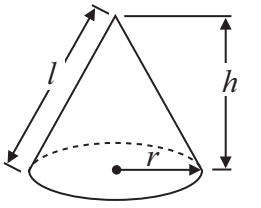
Surface area of sphere = $4\pi r^2$

Volume of sphere = $\frac{4}{3}\pi r^3$



Curved surface area of cone = πrl

Volume of cone = $\frac{1}{3}\pi r^2 h$



The radius of the hemisphere is 3.6 cm.
The radius of the base of the cone is 3.6 cm.
The height of the cone is 12 cm.

Model **A** and model **B** are mathematically similar.

Model **B** is such that
height of model **A** : height of model **B** = 2 : 3

Work out the volume of model **B**.
Give your answer correct to 3 significant figures.

Cone volume = $(1/3)\pi \times 3.6^2 \times 12$

Hemisphere volume = $(2/3)\pi \times 3.6^3$

1 mark
Add the two solid volumes in model A.

A volume = $162.860... + 97.716... = 260.576... \text{ cm}^3$

Length scale = 3/2, so volume scale = $(3/2)^3 = 3.375$

1 mark
Cube the length scale factor.

B volume = $260.576... \times 3.375 = 879.444... \text{ cm}^3$

1 mark
Keep full precision before multiplying.

879 cm^3

1 mark
Round to 3 significant figures.

(Total for Question 16 is 4 marks)

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